B777/B787 Europe (Atlantic) Reference Guide

Flight Planning

- ❑ Download EFLIGHT PLAN (CCI)
- ❑ Evaluate Track Message:
  - check date/time for validity at 30°W
  - check remarks
  - if filed on published track, compare track waypoints to Oceanic Route Verification section of flight plan
  - determine if on a PBCS track
- ❑ Crew passports/visas
- ❑ Review Chart Change Notices/NOTAMS
- ❑ AIREP Form (MISC 84)
- ❑ Download WSI Weather Charts:
  - Wind/Temp Forecast 250mb (FL 340)
  - High Level Sig Wx
- ❑ Volcanic Ash Info
- ❑ Check B777/B787 Fleet Bulletin 26-002: Oceanic Procedures for any updates

Preflight

- ❑ Fit for Duty
- ❑ General Declarations (if required).
- ❑ Customs declarations forms (if required)
- ❑ ETOPS Pre-Departure Check .....Entered in AML
- ❑ Review AML for ETOPS inflight verification check (see Approaching Oceanic Entry Point on page 3).
- ❑ Water/LAV Service.....Verify accomplished
- ❑ Security Search .....Verify accomplished
- ❑ B777 only, SEL 28 (AOM 20.1.1)
  - Aircraft nose numbers: 7AA - 7CA *only*
  - 700 pounds of additional fuel *must* appear on the flight plan when RLS FUEL is greater than 117,000 pounds.
- ❑ NOTOC Acknowledged (if applicable)
- ❑ Verify iPad EFBs current including flight plan
- ❑ Lavatory PA: flights inbound to the US, 'We ask you not to congregate in groups in any area of the cabin especially around the lavatories. Thank you for your cooperation.'"

Enroute

- ❑ Change ICAO STEP on VNAV CRZ page to 1000 feet when evaluating cruise altitudes at OEP (FP, highest acceptable, OPT/MAX/RECMD).
- ❑ Update winds
- ❑ **Logon CPDLC** – when approaching area with CPDLC (10-25 minutes prior to entry).

ACARS Area Weather

ARN	ESSA	Stockholm	SNN	EINN	Shannon
BDA	TXKF	Bermuda	THU	BGTL	Thule
BRU	EBBR	Brussels	WAW	EPWA	Warsaw
CDG	LFPG	Paris	YEG	CYEG	Edmonton
HEL	EFHK	Helsinki	YQX	CYQX	Gander
LGW	EGKK	London Gatwick	YUL	CYUL	Montreal
MAD	LEMD	Madrid	YYC	CYYC	Calgary
MAN	EGCC	Manchester	YYZ	CYYZ	Toronto
OSL	ENGM	Oslo	ZRH	LSZH	Zurich

CPDLC Logon

(Jeppesen Airway Manual - Europe/North America)

BIRD	Reykjavik	EGGX	Shanwick
CDQX	Gander Domestic	EGTT	London
CZEG	Edmonton	EISN	Shannon
CZQM	Moncton	ENOB	Bodo
CZQX	Gander Oceanic	KZWY	New York
CZUL	Montreal	LECB	Barcelona
CZVR	Vancouver	LECM	Madrid
CZWG	Winnipeg	LFFF	Paris
CZYZ	Toronto	LPPC	Portugal
EDUU	Rhein	LPPO	Santa Maria
EDYY	Maastricht	LSAG	Geneva
EGPX	Scottish	LSAZ	Zurich

FIR Communication Formats

Entering a participating flight information region (FIR), contact the FIR radio facility for HF frequency assignments and to establish a SELCAL watch. If the flight will exit the current FIR into another oceanic FIR use the following procedure:
After the radio operator responds, request a SELCAL check and state the next FIR:
Example: "Gander, American ##, SELCAL check, Shanwick next."
If the flight enters an oceanic FIR followed by ATS surveillance airspace (i.e., domestic airspace), follow the procedures described above with the exception that the next FIR should <u>not</u> be stated.
Example: "Shanwick Radio, American ##, SELCAL check."

If the initial call is to a VHF radio operator, they will provide HF frequencies for SELCAL check/watch.
If no answer on first HF call, consider including HF frequency transmitting on in subsequent call.

Santa Maria: Wait to receive the Santa Maria Radio HF frequency assignment via CPDLC.
6 minutes after crossing the FIR boundary, if no uplink HF frequency assignment message is received nor a SELCAL call, advise ATC via CPDLC

Send RCL (Request for Clearance) Message
Gander: 90 - 60 minutes prior to OEP
Reykjavik: 15 minutes prior to OEP
Note: After March 19, RCL message no longer required in Reykjavik Airspace
Santa Maria: at least 40 minutes prior to OEP
Bodo: at least 20 minutes prior to OEP
(See Oceanic Exceptions on page 4.)

For Shanwick (EGGX) Clearance Request Procedures see page 5.

B777-300 and B787
DATALINK Clearance (CZQX, BIRD, LPPO, ENOB)

1. COMM page - select.
2. FLIGHT INFORMATION menu – select.
3. OCEANIC CLEARANCE REQ – select.
4. Enter: FLIGHT NUMBER (if required), ENTRY POINT, ETA, ATC FACILITY (xxxx), FLIGHT LEVEL (3 digits), MACH. Use free text to add "MAX FLXXX" (if required).
5. SEND.

Note: If unable, or requested, revert to voice.

B777-200
RCL Message - Gander OCA

1. COMM page - select COMPANY - FREE TEXT page.
2. "G" for Gander.
3. Line 1 - G/ENTRY POINT/ETA/MACH (no decimals or leading zeros)/FLIGHT LEVEL (2 digits).
   Example line 1: G/TUDEP/1142/84/36
4. Line 2 - MAX FL (3 digits).
   Example line 2: MAX FL380

B777 and B787
After RCL message sent:
First Message- "SENT TO GANDER"(B777-200 *only*)
Second Message- "AALXXX RCL RECEIVED BY (ANSP)"

New York Clearance - Western Atlantic Route System (WAT)
Clearance if transiting New York Oceanic (WAT) into NAT HLA (only from an FAA facility)

1. Clearance received at the departing station (VOICE, DCL or PDC) is a valid clearance through NAT HLA.
2. **No** other route clearance will be issued *unless* there is a change.
3. After ATC requests your FL and Mach, ATC will assign an **altitude** and **Mach** prior to OEP.
   Note: These two items plus the predeparture clearance routing constitute a valid clearance.
4. If cleared via a NAT HLA track, ATC may request (via CPDLC) that you verify your Track Message Identification # (TMI).

The response is given via ATC Free text on CPDLC or voice.

Voice – Eastbound

Voice procedures shall be used to submit an RCL (Request for Clearance) message if:
— Non-ACARS Data Link equipped (777-200)
— ACARS Data Link not operational
— **"RCL REJECTED"** is received by aircraft
— No response to RCL is received within 15 minutes of sending RCL
Automated response to RCL (ACARS or Voice):
"RCL RECEIVED BY [ANSP]"

Gander OCA (CZQX): by VOICE
Gander Oceanic Clearance Delivery not available. Contact domestic controller 200 nm prior to oceanic entry point with RCL message.

Reykjavik OCA (BIRD) – (Eastbound): by VOICE
Note: After March 19, RCL message no longer required. Proceed according to current flight plan.
Transmit RCL message when 20-25 minutes from the Entry Point.
Entry from Edmonton/St. Petersburg: Contact Iceland Radio on HF, or ask domestic controller for RCL message acknowledgment (JeppFD-Pro)

Voice – Westbound

Bodo OACC (ENOB): by voice

1. Contact Bodo on 127.725 not later than 15 minutes prior to entering BODO OACC or
2. BODO Radio on NAT HLA family D frequencies not later than 30 minutes prior to entering. SELCAL check is mandatory.

Reykjavik OCA (BIRD) – (Westbound): Entry over RATSU and points east by VOICE
Note: After March 19, RCL message no longer required. Proceed according to current flight plan.

1. Flights intending to enter NAT HLA Oceanic airspace via RATSU should not call Shanwick with an RCL message. Call Iceland radio on VHF primary on 127.85, secondary 129.625 to send the RCL message.
2. The RCL message acknowledgment will be issued by Reykjavik Control. There are three points established at the boundary of delegated airspace from Scottish to Reykjavik, BESGA, DEVBI and BARKU on routes to RATSU.
3. Air crews approaching Reykjavik airspace from the Scottish and Stavanger areas can contact Iceland Radio on VHF primary 127.85, secondary 129.625 to relay the RCL message.

Santa Maria OACC (LPPO): Voice or Data Link - at least 40 minutes prior to OCA

1. Call on VHF frequency 132.07/ HF 8906 and transmit RCL message. Give: Entry Point and Time / Altitude and Mach Requested and SELCAL.
2. Standby on VHF or HF SELCAL for acknowledgment of RCL message.

Voice RCL Message and Read Back Formats

RCL Message

"American XX estimating ___ (entry fix) at ___Z, requesting FL___, Mach___" (And highest acceptable FL which can be maintained by OCA entry point).

Response - RCL message received by (ANSP)

If ATC issues a route amendment:

- ❑ PF and PM will separately review the amended clearance vs FMC
- ❑ PM will compare amended clearance to FMC LEGS (expand all latitude/longitude waypoints) and note any changes
- ❑ PF will compare amended clearance to FMC LEGS (expand all latitude/longitude waypoints); any differences will be immediately changed in the FMC by the PF while the PM confirms the accuracy of the changes
- ❑ Advise dispatch of revised routing
- ❑ Obtain new Critical Fuel Scenario and Oceanic Route Verification
- ❑ Obtain new FLT PROGRESS/FPR
- ❑ One pilot (PM or PF) will read aloud FMC waypoints while the other pilot checks them against the amended clearance
- ❑ Do not respond to CPDLC CONFIRM ASSIGNED ROUTE message until all route amendments are entered in the FMC, confirmed, and executed
- ❑ Update winds for new route/altitude

If no route amendment is issued by ATC:

- No further legs check is required if a successful preflight route and legs check has been accomplished

Note: Half-degree coordinates can only be checked by expanding the waypoints. Caution is required to ensure these coordinates are correctly entered in the FMS.

Prior to Oceanic Entry Point (OEP)

- ❑ If required by a TAC item, determine the ETOPS entry/exit point using the fix page or lat/long from the Critical Fuel Scenario.

ETOPS Alternate Planning/Diversion Guide:

- ❑ Update alternate WX, forecast, and NOTAMs.
- ❑ Evaluate potential diversion airport(s): **FIRE**, Mechanical, Medical/Passenger.

Approaching Oceanic Entry Point

- ❑ Check ECON speed (cost index) or assigned MACH
- ❑ Confirm at cleared cruise altitude and set in FMC
- ❑ If required, complete any **ETOPS In Flight Verification Check(s)**; such as an in flight engine check, an in flight APU start, etc., prior to ETOPS entry.

Note: Ensure the required AML entries are made prior to entering ETOPS airspace.

After Oceanic Entry Point

- ❑ Operational HF/SELCAL check (if not already done).
- ❑ Tune left VHF radio to 121.5 and right VHF to 123.45.
- ❑ Transponder to 2000 when 10 minutes past oceanic entry or as directed (REMINDERS page).
- ❑ Establish SLOP in all NAT/HLA [see Strategic Lateral Offset Procedure(SLOP) on page 4].
- ❑ Update winds for new route/altitude.

CPDLC Messages

1. Set max uplink delay to 300 seconds or requested value.
   — Select ATC from Communication Main Menu.
   — Select LOGON/STATUS.
   — Set MAX UPLINK DELAY to 300 seconds or requested value.
2. Confirm assigned route (see Confirm Assigned Route on page 4 for more information).
3. Resume normal speed (see Resume Normal Speed on page 4 for more information).

Oceanic Procedures

Approaching Active Oceanic Waypoint

Both pilots must verify that the expanded full Lat/Long coordinates for the next and the following waypoint agree with the active flight plan and/or ATC clearance, as well as the Oceanic Route Verification.

Waypoint Passage

1. Confirm next waypoint becomes active waypoint.
2. Verify autopilot is coupled to LNAV.
3. Record time and fuel on FPR.
4. Complete AIREP Form/transmit report (if required).
5. Evaluate potential diversion airport(s): fire, mechanical, medical/passenger

Midpoint Check

1. Check ETA for active waypoint. If ETA has changed by 3 minutes or more from what was reported to ATC revise estimate using CPDLC, ATC free text, or voice. If transmitting position via ADS-C, no need to update ETA.
2. Check for satisfactory fuel quantity/balance and trend.

Oceanic Exit

1. Confirm ATC clearance (if required).
2. If applicable, communicate acceptance/rejection of redispatch message. Redispatch may be received up to 2 hours prior; however, if not received within 1 hour from redispatch point, contact the dispatcher.
3. Request desired altitude and Mach from ATC.
4. Confirm SLOP offset is zero.

Note: Do not go direct to the oceanic exit point.

5. Change STEP on VNAV CRZ page to 2000 feet to evaluate cruise altitudes after oceanic exit (FP, highest acceptable, OPT/MAX/RECMD)

Standard Entry/Exit Briefing

Prior to PF or captain leaving the flightdeck or after any assigned pilot re-enters the flightdeck while enroute:

- ❑ PF brief/PM review aircraft status from FMA.
- ❑ PF brief additional items (aircraft systems status, weather, ATC clearance/frequency change).
- ❑ For a crew rest break the captain/PIC will brief:
  - a designated PIC
  - weather deviation options and procedures
  - current and *expect* clearances (e.g., redispatch, step climbs, etc.)
  - diversion options to include alternate airports and weather, departure from track/airway procedures, ETPs, and terrain awareness areas
  - any other operational considerations (e.g., datalink clearances, required radio frequency or transponder code changes)
  - crew rest location(s) (bunk or rest seat) as applicable

SATCOM Quick Reference

For ATC communications CPDLC/HF may be more effective.

SATCOM INMARSAT (B777)

1. MENU, SAT (2L), AA DISP #s [5R].
2. Select Dispatch Desk #.
3. For INMARSAT select ATC emergency from SAT MAIN MENU scratch pad, enter in 1L or 3L.
4. PLACE CALL (2R or 4R).

SATCOM INMARSAT (B787)

1. Select SAT on the TCP.
2. From the SAT PHONE page, if the desired dispatch desk (DSP DSK XX) or ground station is displayed in line 1L or 3L, proceed to step 4.
3. To select a ground station from the SAT DIRECTORY, select DIRECTORY at line 2R. Select the desired ground station to SAT 1 or SAT 2. The display will return to the SAT PHONE page.
4. If required, select a priority (EMG, HGH or LOW). Press the MAKE CALL for the desired SAT radio.

Strategic Lateral Offset Procedure(SLOP)

SLOP is a standard operating procedure for the entire region and applies for organized tracks (OTS) and random routes. The options are a 1 or 2 nm offset to the right, or centerline.

Note: Aircraft (e.g., B787) able to perform offsets in tenths of nautical mile should do so as it contributes to risk reduction.

1. Offset using LNAV, not heading select.
2. Do not advise ATC or request ATC clearance.

Return to course by oceanic exit point by deleting offset.

Confirm Assigned Route

- ❑ Select DISPLAY REPORT prompt.
- ❑ ATC REPORT page shows SLOP as DEV.
- ❑ Select SEND.

AIREP Form (FO-0024)

Use the AIREP form when:

— operating in (either the North Atlantic or Pacific oceanic airspace) and non-radar position reporting is required

— in other areas as desired

Note: Available via ACARS - MISC 84

Resume Normal Speed

Remove the speed restriction:

If the aircraft receives RESUME NORMAL SPEED (via CPDLC or voice), there is no longer a need to comply with a previously issued Mach. Advise ATC if speed will be adjusted plus or minus Mach 0.02 or more from filed cost index Mach.

Oceanic Exceptions

New York

If a flight enters New York oceanic at SLATN, JOBOC, or DOVEY **and** does not enter airspace delegated to either Moncton or Gander ACC, then fly the clearance received via CPDLC-DCL *unless* amended. New York ATC will confirm requested altitude and speed. The TMI is required during the readback if assigned an organized track.

Moncton

If a flight enters airspace delegated to Moncton ACC **and** returns to New York oceanic (via AVAST, NOVOK, or JEBBY) **and** does not enter Gander Domestic ACC airspace, then fly the clearance received via CPDLC-DCL *unless* amended. Moncton ATC will confirm requested altitude and speed.

Gander

If a flight enters airspace delegated to Gander ACC (DOPHN, JAROM, BOBTU, TALGO, or RAFIN) via Moncton ACC or New York oceanic **and** enters NAT airspace through either Gander or New York then New York then send RCL message to Gander oceanic via ACARS (data link) or voice.

Loss of Cabin Pressure

See AOM 17.6 Emergency Descent. This procedure is in ECL> Unannunciated Checklists and QRH> Miscellaneous. Turn toward a suitable airfield when appropriate.

Medical

1. Refer to FOM> Fast Reference Links> Medical Emergencies.
2. If Big 4, diversion required: notify Captain, Dispatch. Obtain revised clearance. If unable, accomplish Departure from Route/Track on page 7

Shanwick Clearance Request Procedures

Shanwick Oceanic Clearance

1. **A separate clearance is required prior to entry into Shanwick North Atlantic High Level Airspace (NAT/HLA).**
2. Use one of the following options to get an Oceanic Clearance: DATA LINK, ACARS, or VOICE.
   - Data Link: B777-300, B787
   - COMM/COMPANY/Free Text: B777-200
   - If unable DATA LINK/COMPANY: revert VOICE
3. Obtain Oceanic Clearance 90-30 minutes prior to OEP
4. Use same procedures for transmitting RCL messages to request Oceanic Clearance from Shanwick

B777-200
RCL Message - Shanwick OCA

1. COMM page - select COMPANY - FREE TEXT page.
2. "S" for Shanwick.
3. Line 1 - S/ENTRY POINT/ETA/MACH (no decimals or leading zeros)/FLIGHT LEVEL (2 digits).
   Example line 1: S/TUDEP/1142/84/36
4. Line 2 - MAX F (3 digits).
   Example line 2: MAX FL380

B777 and B787

After oceanic clearance request sent:

First Message: "SENT TO SHANWICK" (B777-200)

Second Message: "AALXXX RCL RECEIVED"

Westbound: "IF NO CLEARANCE RECEIVED AFTER 15 MINUTES REVERT TO VOICE"

Third Message: "ATC OCEANIC CLEARANCE"

COMM alert COMM SWITCH - Press: SELECT PRINT

Press ACPT (accept) on Glareshield Panel

Fourth Message: "CLEARANCE CONFIRMED"

Shanwick OCA (EGGX): by voice (Data Link not available)

1. Contact Shanwick Oceanic Clearance Delivery on 123.95 anytime within VHF range.
2. Relay RCL message using clearance request format (see page 3).
3. Read back RCL message acknowledgment.

DEL, DOH and TLV Westbound Flights

In remarks section add RNP capability (RNP 4 or RNP 10)

Voice Clearance Request and Read Back Formats

Clearance Request

"American XX requesting oceanic clearance, estimating ___ (entry fix) at ___Z, requesting FL___, Mach___" (And highest acceptable FL which can be maintained by OCA entry point).

Read Back – NAT HLA Clearance

"American ## is cleared via track__, TMI___, FL___, Mach___".

Read Back – Random Route or "Via Flight Planned Route"

"American # is cleared via, <fix>, <fix>, <fix>, <fix>, etc. Mach____, FL_____."

Note: If any doubt, give a full readback rather than an abbreviated one.

After Receipt of Oceanic Clearance

Fly the **clearance** not the flight plan (FP)

Accomplish the following clearance check:

PF and **PM**- separately review the oceanic clearance vs FMS.

- ❑ **PM** will compare the oceanic clearance to the FMS LEGS (expand all oceanic waypoints) and note any changes.
- ❑ **PF** will compare the oceanic clearance to the FMS LEGS (expand all oceanic waypoints); any differences will be immediately changed in the FMS by the **PF** while the **PM** confirms the accuracy of the changes

Note: In oceanic airspace, fly clearance as received until ATC issues further routing. Planned routing (from dispatch) should not be construed as an ATC clearance.

If Clearance Differs from Flight Plan (Re-Route)

- ❑ Advise dispatch of revised routing.
- ❑ Obtain new flight plan via FREE TEXT or correct current MFP.
- ❑ Obtain revised FLT PROGRESS/FPR and Oceanic Route Verification.
- ❑ Update winds for new route/altitude.

Note: In oceanic airspace, fly clearance as received until ATC issues further routing. Planned routing (from dispatch) should not be construed as an ATC clearance.

Confirm master flight plan and oceanic clearance agree.

Note: Continue with "Prior to Oceanic Entry Point (OEP)" on page 3.

Weather Deviations

Weather Deviations

Revised Clearance

If a lateral deviation from a route/track is necessary to avoid weather (e.g., thunderstorms), request a revised ATC clearance as soon as possible by stating “Weather deviation required.” or by a CPDLC lateral down link message.

1. Select “COMM MENU”
2. Select “ATC”
3. Select “ROUTE REQUEST”, then “WX DEVIATION”
4. Enter requested deviation mileage and direction. If no direction is entered, controller will assume requested deviation is left and right of course.

Obtain essential traffic information prior to deviating; even for a small distance (e.g., less than 5 nm). If a revised ATC clearance cannot be obtained, follow the procedures described below.

After the deviation is started, continue to obtain a revised ATC clearance.

Deviation Beyond ATC Clearance Parameters

ATC can only provide a revised clearance which will continue to assure minimum standard separations are maintained from all other traffic, including any adjacent routes/tracks.

When this is not possible, ATC will advise “Unable and state intentions.”

The pilot should indicate the direction, anticipated magnitude and, if appropriate, the expected altitude change of the intended deviation.

Deviation with ATC Clearance

If a revised clearance is received, it should be followed in all its provisions.

If such a revised clearance is received after beginning the contingency procedures, the revised clearance supersedes the contingency procedure provisions.

Deviation without ATC Clearance

If revised ATC clearance has not been obtained, use the following procedures:

1. Deviate away from the organized track or route system, if possible.
2. Establish communications with and alert nearby aircraft broadcasting, at suitable intervals:
   - aircraft identification, flight level, aircraft position (including ATS route designator or the track code) and intentions.

   Broadcast on the frequency in use (when VHF) and on frequency 121.5 MHz (or, as a back-up, on the VHF inter-pilot air-to-air frequency 123.45 MHz) or use CPDLC.

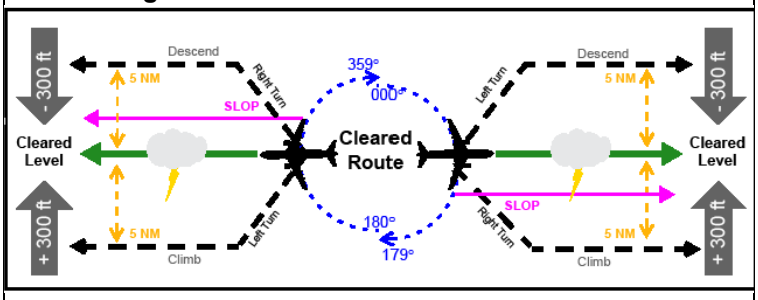
   Note: When necessary use the urgency call “pan-pan” or the CPDLC urgency down link message.
3. Watch for conflicting traffic both visually and by reference to TCAS.
4. Turn on all aircraft exterior lights.
5. For deviations of:
   - **less than 5** nautical miles:
     remain at the ATC assigned flight level
   - **equal to or greater than 5** nautical miles:
     when the aircraft is 5 nautical miles or greater from the magnetic track/route centerline, initiate a level change of 300 feet

   Note: Adjust the aircraft path to avoid conflict with other aircraft at or near the same flight level.

   When the magnetic track/route centerline is:

   — east 000° - 179° and you are:
   - **left** of course: descend 300 feet
   - **right** of course: climb 300 feet

   — west 180° - 359° and you are:
   - **left** of course: climb 300 feet
   - **right** of course: descend 300 feet



Example: Your magnetic track/ route centerline is 175° and you are deviating 5 nm (and possibly more) to the right of your course: you need to initiate a 300 foot climb when passing 5 nm.

When returning to route/track:

6. Resume the last assigned flight level when the aircraft is less than 5 nm from the centerline.

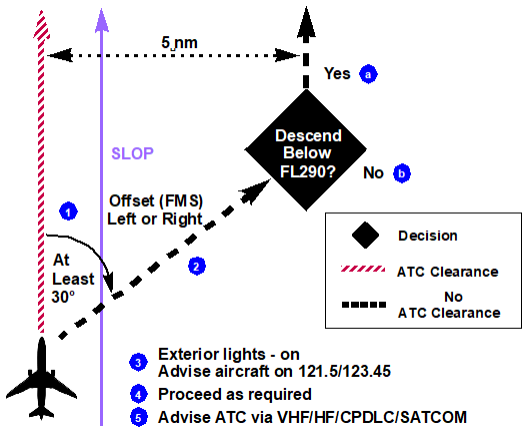
Deviation Complete

Advise ATC when the weather deviation is no longer required, or when a weather deviation has been completed and the aircraft has returned to the centerline (or previously adopted SLOP offset) of its cleared route.

Note: Weather deviations beyond 128 nm are not possible due to FANS limitations. If more than 128 nm is needed, the flight crew must submit a Route Request containing a waypoint that will avoid the weather. Include in the Route Request a downtrack waypoint on the flight plan, past the weather.

Departure from Route/Track

Departure from Route/Track



If unable to continue flight in accordance with an ATC clearance due to inflight contingencies (equipment failure, etc.) or an emergency, obtain a revised clearance, whenever possible, prior to initiating action. Contact ATS via CPDLC/VHF/HF/SATCOM, or request dispatch via SATCOM to contact ATS, to advise of actions and request clearance (time and conditions permitting).

Unable to Obtain Clearance

If unable to obtain an ATC clearance in a timely manner, the captain should exercise emergency authority by executing whatever action necessary to ensure flight safety.

Note: The aircraft should be flown at a flight level and altitude where other aircraft are less likely to be encountered.

Use the following procedures (also located on the specific-region reference guides) until a revised clearance is received:

1. Leave the assigned route or track by turning at least 30° to the left or right (offset/LNAV) based on:
   - aircraft position relative to any organized track or route system,
   - direction of flights and flight levels allocated on adjacent tracks,
   - direction to an alternate airport,
   - any strategic lateral offset being flown, and
   - terrain clearance.
2. Maintain the assigned flight level (if possible) until established on the 5 nm parallel, same direction track/route offset.

   Note: If unable to maintain the assigned flight level, minimize climb/descent rate to the extent that is operationally feasible.

   Once established on a 5 nm parallel, same direction track/route offset, decide whether to continue descent to below FL290.

   Descent below FL290 is particularly applicable to operations where there is a predominant traffic flow (e.g., east-west) or parallel track system where the aircraft's diversion path will likely cross adjacent tracks or routes. Descending below FL290 can decrease the likelihood of conflicts with other aircraft, TCAS RA events, and delays in obtaining a revised ATC clearance.

   a. If continuing the descent to below FL290 and once below FL290:

      establish/maintain an altitude that differs by 500 feet from altitudes normally used.

   b. If <u>not</u> continuing the descent:

      establish/maintain an altitude that differs by 500 feet (1000 feet if above FL410) from altitudes normally used.

   Note: Altimetry system error may lead to less than actual 500 feet vertical separation and TCAS RAs may occur.
3. Turn on all exterior aircraft lights (IAW aircraft operating manual limitations) and broadcast position and intentions at suitable intervals on appropriate air-to-air, air-to-ground frequencies and 121.5 MHz to alert other aircraft in the area using “pan-pan” (urgent) or “mayday” (distress) as appropriate. Proceed as required by the operational situation or if an ATC clearance has been obtained, proceed in accordance with the clearance.
4. Advise ATC: “mayday” or “pan-pan” (as appropriate) using CPDLC, SATCOM, or 121.5.
5. Record/annotate time and approximate position for any reports that may be required.
6. Submit appropriate pilot debrief (if necessary).

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